TCS Coding Practice Qs

How to Prepare for Coding

- DSA Questions – Medium Level (Leetcode)
  - Arrays
  - Strings
  - Recursion
  - Linked List
  - Greedy Algorithm

Q - Move Empty Chocolate Packets to the End

Problem Description:
A chocolate factory is packing chocolates into packets.
Each packet is represented by an integer value in an array.
A value of 0 represents an **empty packet**, while a non-zero value represents a **packet with chocolates**.
Your task is to **move all the empty packets (0s) to the end of the array**, while maintaining the **order of the non-empty packets**.
You must perform this operation **in-place** (i.e., do not use extra arrays).

Sample Input:
8 4 5 0 1 9 0 5 0

Sample Output:
4 5 1 9 5 0 0 0

Q - Move Empty Chocolate Packets to the End

```
def move_empty_packets(arr):
    n = len(arr)
    count = 0  # Tracks index for non-zero elements
    for i in range(n):
        if arr[i] != 0:
            arr[count] = arr[i]
            count += 1

    while count < n:
        arr[count] = 0
        count += 1
    return arr

# Driver Code
N = int(input())
arr = list(map(int, input().split()))
result = move_empty_packets(arr)
print(*result)
```

Q - Find Non-Repeating Elements

- **You are given an array of integers.**
  Your task is to find and print all the non-repeating elements (i.e., elements that appear exactly once) from the array.

```
def find_non_repeating_elements(arr):
    freq = {}
    # Count frequency of each element
    for num in arr:
        freq[num] = freq.get(num, 0) + 1
    # Collect elements with frequency 1 (non-repeating)
    result = [num for num in arr if freq[num] == 1]
    return result

# Driver Code
N = int(input())   #10
arr = list(map(int, input().split()))    #1 2 3 2 1 4 5 6 5 7
result = find_non_repeating_elements(arr)
print(result)
```